

z2 (5.6)

Rozwiąż na $[-\pi, \pi]$:

$$\sqrt{3} \cos^2 x - 2 \sin x \cos x - \sqrt{3} \sin^2 x = 0.$$

① Sprawdzamy dla $\cos x = 0$.

$$\sqrt{3} \cdot 0 - 2 \cdot 1 \cdot 0 - \sqrt{3} \cdot 1 = 0$$

$$-\sqrt{3} = 0$$

sprzeczność

② Dzielimy przez $\cos^2 x \neq 0$

$$\sqrt{3} \frac{\cos^2 x}{\cos^2 x} - 2 \frac{\sin x \cos x}{\cos^2 x} - \sqrt{3} \frac{\sin^2 x}{\cos^2 x} = 0$$

$$\sqrt{3} - 2 \operatorname{tg} x - \sqrt{3} \operatorname{tg}^2 x = 0$$

$$\sqrt{3} \operatorname{tg}^2 x + 2 \operatorname{tg} x - \sqrt{3} = 0$$

③ Korzystamy z pomocniczej zmiennej.

$$t = \operatorname{tg} x$$

$$\sqrt{3} t^2 + 2t - \sqrt{3} = 0$$

$$\Delta = 4 + 4 \cdot \sqrt{3} \cdot \sqrt{3} = 16 \quad \sqrt{\Delta} = 4$$

$$t_1 = \frac{-2-4}{2\sqrt{3}} = \frac{-3}{\sqrt{3}} = -\sqrt{3} \quad t_2 = \frac{-2+4}{2\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{3}$$

④ Łafamy podstawienie.

$$\operatorname{tg} x = -\sqrt{3} \quad \operatorname{tg} x = \frac{\sqrt{3}}{3}$$

$$x = -\frac{\pi}{3} + k\pi \quad x = \frac{\pi}{6} + k\pi, \quad k \in \mathbb{Z}$$

$$\begin{array}{lll} \text{Dla } k=0 & x = -\frac{\pi}{3} & x = \frac{\pi}{6} \\ \text{Dla } k=1 & x = \frac{2\pi}{3} & x = \frac{7\pi}{6} \notin [-\pi, \pi] \\ \text{Dla } k=-1 & x = -\frac{4\pi}{3} \notin [-\pi, \pi] & x = -\frac{5\pi}{6} \end{array}$$

$$\text{Odp. : } x \in \left\{ -\frac{\pi}{3}, \frac{2\pi}{3}, \frac{\pi}{6}, -\frac{5\pi}{6} \right\}.$$